

!TOC SQL Employee Famous Question

SQL Employee — Interview Revision

Last Synced : 2026/10/05

- Delete duplicate records while keeping one record.
- Find departments having more than 5 employees, with department name.
- Find duplicate employee names.
- Find duplicate records.
- Find duplicate salaries.
- Find employees having the same department and salary.
- Find employees having the same manager.
- Find employees who don't have a manager.
- Find employees who earn more than their manager.
- Find employees who joined before their manager.
- Find employees who joined in the last 30 days.
- Find employees whose salary is greater than the average salary.
- Find employees whose salary is greater than their department's average salary.
- Find employees with the same salary.
- Find managers managing more than 5 employees.
- Find the 2nd highest salary in each department.
- Find the 2nd highest salary.
- Find the Nth highest salary in each department.
- Find the Nth highest salary without LIMIT/TOP.
- Find the Nth highest salary.
- Find the department with the highest average salary.
- Find the highest salary in each department.
- Find the maximum salary without using MAX().
- Find the salary difference between employee and manager.
- Find the second highest salary in each department.
- Find the top 3 salaries in each department.
- Finds Managers Who Have More Than One Employee.

Pattern coverage

Salary / Ranking

- └─ 2nd highest
- └─ Nth highest
- └─ Nth highest without LIMIT
- └─ Top N per department
- └─ Nth highest per department

GROUP BY / HAVING

- └─ Duplicate records
- └─ Duplicate salary

- └─ Same dept + salary
- └─ > 5 employees
- └─ Highest average salary

SELF JOIN

- └─ Employee > Manager
- └─ Joined before Manager
- └─ Same Manager
- └─ Salary difference

DATE

- └─ Joined in last 30 days

DUPLICATES

- └─ Find duplicates
- └─ Delete duplicates

Create And Praticce it.

sql

```
CREATE DATABASE IF NOT EXISTS sql_pratice_2026;  
USE sql_pratice_2026;
```

```
DROP TABLE IF EXISTS employee;
```

```
CREATE TABLE employee (  
  employee_id INT PRIMARY KEY,  
  employee_name VARCHAR(100),  
  salary INT,  
  dept_id INT,  
  manager_id INT,  
  joining_date DATE  
);
```

```
INSERT INTO employee  
(employee_id, employee_name, salary, dept_id, manager_id, joining_date)  
VALUES  
(1, 'Amit', 100000, 10, NULL, '2020-01-10'),  
(2, 'Rahul', 80000, 10, 1, '2021-03-15'),  
(3, 'Priya', 80000, 10, 1, '2022-06-20'),  
(4, 'Neha', 70000, 20, NULL, '2020-05-12'),  
(5, 'Vikas', 60000, 20, 4, '2023-02-10'),  
(6, 'Sneha', 50000, 20, 4, '2024-01-05'),  
(7, 'Arjun', 90000, 30, NULL, '2021-07-18'),  
(8, 'Karan', 75000, 30, 7, '2025-09-10'),  
(9, 'Pooja', 75000, 30, 7, '2026-09-20'),  
(10, 'Amit', 60000, 20, 4, '2026-09-25');
```

Find the second highest salary in each department.

use sql_practice_2026;

-- Date: Oct 3, 2026 | Time: 11:06:41 PM | MYSQL 8.0+

-- SQL Question : Find the second highest salary in each department.

SELECT * FROM employee ;

SELECT

FROM (

SELECT

e,

DENSE_RANK() OVER(PARTITION BY dept_id ORDER BY salary DESC) AS rnk

FROM employee e

) x

WHERE rnk=2;

Delete duplicate records while keeping one record

use sql_practice_2026;

-- Date: Oct 5, 2026 | Time: 1:09:55 PM | MYSQL 8.0+

-- SQL Question : Delete duplicate records while keeping one record

SELECT * FROM employee;

-- DELETE On the basis of what ?

-- How to find duplicate Record ?

SELECT dept_id,COUNT() AS cnt FROM employee GROUP BY dept_id HAVING COUNT()>1;

-- ❌ DELETE FROM employee IN (SELECT dept_id,COUNT() AS cnt FROM employee GROUP BY dept_id HAVING COUNT()>1); -- What is the issue with this code ?

-- ❌ DELETE FROM employee IN (SELECT dept_id FROM employee GROUP BY dept_id HAVING COUNT()>1); -- What is the issue with this code ? | What about now ?

-- ❌ DELETE FROM employee WHERE dept_id IN (SELECT dept_id FROM employee GROUP BY dept_id HAVING COUNT()>1); -- SQL Error [1093]

-- ❌ DELETE FROM employee WHERE dept_id IN (SELECT MIN(dept_id) FROM employee GROUP BY dept_id HAVING COUNT(*)>1);

-- USE RANK Much Better And HOW ?

GROUP BY dept

Order IDK

Find departments having more than 5 employees.

use sql_practice_2026;

-- Date: Oct 3, 2026 | Time: 9:12:44 PM | MYSQL 8.0+

-- SQL Question : Find departments having more than 5 employees.

SELECT * FROM employee;

SELECT dept_id,COUNT(*) AS cnt FROM employee GROUP BY dept_id HAVING COUNT(*)>5; -- 1,2,3,4,5,6

-- If I need to display name as well ?

-- SELECT dept_id,employee_name FROM employee GROUP BY dept_id HAVING COUNT(*)>5;

-- What is the issue with this line of code ?

-- So MySQL with ONLY_FULL_GROUP_BY enabled will reject it.

SELECT FROM employee

WHERE dept_id IN (

SELECT dept_id FROM employee GROUP BY dept_id HAVING COUNT(*)>5

) ORDER BY dept_id;

Find the top 3 salaries in each department.

use sql_practice_2026;

-- Date: Oct 3, 2026 | Time: 6:28:36 PM | MYSQL 8.0+

/* Possible Approaches :

*/

-- SQL Question Find the top 3 salaries in each department.:

-- GROUP BY | Reason : It combine the result

SELECT dept_id,MAX(salary) AS salary FROM employee

GROUP BY dept_id

ORDER BY

dept_id ASC,

salary DESC ;

-- MAX() → gives only the highest salary,

-- not the top 3 salaries.

-- Can we solve by GROUP ? NOOOOOOOO

-- Approach 2: DENSE_RANK()

SELECT FROM

(

SELECT

e,

```
DENSE_RANK() OVER(PARTITION BY dept_id ORDER BY salary DESC) AS rnk  
FROM employee e  
) x  
WHERE rnk<=3  
ORDER BY dept_id ASC, salary DESC;
```

Find duplicate employee names

use sql_practice_2026;

-- Date: Oct 5, 2026 | Time: 1:40:23 PM | MYSQL 8.0+

-- SQL Question : Find duplicate employee names

```
SELECT  
employee_name,  
COUNT() AS cnt  
FROM employee  
GROUP BY employee_name  
HAVING COUNT()>1;
```

Find duplicate records

use sql_practice_2026;

-- Date: Oct 3, 2026 | Time: 11:26:20 PM | MYSQL 8.0+

-- SQL Question : 3. Find duplicate records.

SELECT * FROM employee;

-- How do you define duplicate record

-- Suppose for sake I consider employee_name

SELECT e1.*

FROM employee e1

JOIN employee e2

ON e1.employee_name = e2.employee_name

WHERE e1.employee_id != e2.employee_id;

Find duplicate salaries.

use sql_pratice_2026;

-- Date: Oct 3, 2026 | Time: 11:34:54 PM | MYSQL 8.0+

-- SQL Question : Find duplicate salaries.

SELECT salary FROM employee GROUP BY salary HAVING COUNT(*)>1;

SELECT *FROM employee*

WHERE salary IN (

SELECT salary FROM employee GROUP BY salary HAVING COUNT()>1

);

Find employees having the same department and salary.

use sql_practice_2026;

-- Date: Oct 5, 2026 | Time: 12:21:06 AM | MYSQL 8.0+

-- SQL Question : Find employees having the same department and salary.

SELECT * FROM employee;

-- JOIN

SELECT

e1.employee_id,

e1.employee_name,

e1.salary,

e2.employee_id,

e2.employee_name,

e2.salary

FROM employee e1

JOIN employee e2

ON e1.dept_id = e2.dept_id

AND e1.salary = e2.salary

AND e1.employee_id <> e2.employee_id -- CAN WE USE DISTINCT ??

ORDER BY e1.employee_name;

-- Should we use DISTINCT ?

Find employees having the same manager.

```
use sql_practice_2026;
```

```
-- Date: Oct 5, 2026 | Time: 1:42:17 PM | MYSQL 8.0+
```

```
-- SQL Question : Find employees having the same manager.
```

```
SELECT * FROM employee;
```

```
-- I need to find e1, e2 that had manager_id as m1
```

```
-- SELF JOIN
```

```
SELECT
```

```
e1.employee_id,
```

```
e1.employee_name,
```

```
e1.manager_id,
```

```
e2.employee_id,
```

```
e2.employee_name,
```

```
e2.manager_id
```

```
FROM employee e1
```

```
JOIN employee e2
```

```
ON e1.manager_id = e2.manager_id
```

```
AND e1.employee_id < e2.employee_id; -- Do i really need to avoid duplicate
```

```
-- Will <> Resolve the issue ?
```

```
-- 101 → 102
```

```
-- 102 → 101 ← duplicate/reverse pair
```

```
-- <> → avoids self-pair
```

```
-- < → avoids self-pair + reverse duplicate
```

Find employees who don't have a manager

use sql_practice_2026;

-- Date: Oct 3, 2026 | Time: 9:30:39 PM | MYSQL 8.0+

-- SQL Question : Find employees who don't have a manager

SELECT * FROM employee;

-- Simplest Approach

SELECT *

FROM employee

WHERE manager_id IS NULL;

-- JOIN Approach

SELECT *

FROM employee e

LEFT JOIN employee m -- Why Left Join ?

ON e.manager_id = m.employee_id

WHERE e.manager_id IS NULL;

Find employees who earn more than their manager

use sql_practice_2026;

-- Date: Oct 3, 2026 | Time: 8:26:35 PM | MYSQL 8.0+

/* Possible Approaches :

*/

-- SQL Question : Find employees who earn more than their manager

SELECT

e.employee_name,

e.salary

FROM employee e

JOIN employee m

ON m.employee_id = e.manager_id

WHERE e.salary > m.salary

Find employees who joined before their manager

use sql_practice_2026;

-- Date: Oct 5, 2026 | Time: 1:56:03 PM | MYSQL 8.0+

-- SQL Question : Find employees who joined before their manager.

SELECT

e.employee_name,

e.joining_date,

m.employee_name,

m.joining_date

FROM employee e

JOIN employee m

ON e.manager_id = m.employee_id

WHERE e.joining_date < m.joining_date;

-- JOIN

Find employees who joined in the last 30 days

use sql_practice_2026;

-- Date: Oct 3, 2026 | Time: 11:09:45 PM | MYSQL 8.0+

-- SQL Question : Find employees who joined in the last 30 days.

SELECT * FROM employee WHERE joining_date >= '2026-09-03'; -- Ignore it.

SELECT *FROM employee WHERE DATEDIFF(CURRENT_DATE,joining_date) BETWEEN 0 AND 30;*

SELECT FROM employee WHERE joining_date >= CURRENT_DATE - INTERVAL 30 DAY;

Find employees whose salary is greater than the average salary

use sql_practice_2026;

-- Date: Oct 3, 2026 | Time: 6:36:19 PM | MYSQL 8.0+

/* Possible Approaches :

*/

-- SQL Question Find employees whose salary is greater than the average salary:

SELECT * FROM employee

WHERE salary > (

SELECT AVG(salary)

FROM employee

);

Find employees whose salary is greater than their department's average salary

use sql_practice_2026;

-- Date: Oct 5, 2026 | Time: 2:20:23 PM | MYSQL 8.0+

-- SQL Question : Find employees whose salary is greater than their department's average salary

SELECT * FROM employee;

--  SELECT * FROM employee WHERE salary > (SELECT AVG(salary) AS salary FROM employee GROUP BY salary);

SELECT * FROM employee e

WHERE e.salary > (

SELECT AVG(e2.salary)

FROM employee e2

WHERE e2.dept_id = e.dept_id

);

Find employees with the same salary.

use sql_practice_2026;

-- Date: Oct 3, 2026 | Time: 9:21:55 PM | MYSQL 8.0+

-- SQL Question : Find employees with the same salary.

-- Thinking of Self Join

SELECT

e1.employee_name,

e1.salary,

e2.employee_name,

e2.salary

FROM

employee e1

JOIN employee e2

ON e1.salary = e2.salary -- Why on salary ? || Whyn't on ID ?

WHERE e1.employee_id > e2.employee_id; -- To Mantain Uniqueness

Find managers managing more than 5 employees

use sql_pratice_2026;

-- Date: Oct 5, 2026 | Time: 2:31:59 PM | MYSQL 8.0+

-- SQL Question : Find managers managing more than 5 employees

SELECT * FROM employee;

```
SELECT
manager_id,
COUNT() AS cnt
FROM employee
GROUP BY manager_id
HAVING COUNT()>5;
```

Find the 2nd highest salary in each department

use sql_practice_2026;

-- Date: Oct 5, 2026 | Time: 2:43:42 PM | MYSQL 8.0+

-- SQL Question : Find the 2nd highest salary in each department

-- Thinking of DENSE RANK And Partition

SELECT *FROM* (

SELECT

e,

DENSE_RANK() OVER(PARTITION BY dept_id ORDER BY salary DESC) AS rnk

FROM employee e

) x

WHERE rnk=2

Find the 2nd highest salary

use sql_pratice_2026;

-- Date: Oct 3, 2026 | Time: 12:04:41 PM | MYSQL 8.0+

/* Possible Approaches :

*/

-- SQL Question : Find the 2nd highest salary.

-- SELECT DISTINCT salary FROM employee ORDER BY salary DESC OFFSET 1 LIMIT 1; SQL Version

SELECT DISTINCT salary FROM employee ORDER BY salary DESC LIMIT 1, 1;

SELECT MAX(salary) AS second_highest_salary

FROM employee e

WHERE e.salary < (

SELECT MAX(salary) FROM employee

);

Find the department with the highest average salary

use sql_practice_2026;

-- Date: Oct 4, 2026 | Time: 11:17:13 PM | MYSQL 8.0+

-- SQL Question : Find the department with the highest average salary

-- Step 1: Find average salary of each department

```
SELECT
dept_id,
AVG(salary)
FROM
employee
GROUP BY
dept_id;
```

-- Step 2: Find the maximum average

-- Derived Table Or Inline table | VALID ANSWER

```
SELECT MAX(avg_salary) FROM
(
SELECT
dept_id,
AVG(salary) as avg_salary
FROM
employee
GROUP BY
dept_id
) t;
```

```
SELECT
dept_id,
AVG(salary) AS avg_salary
FROM employee
GROUP BY dept_id
ORDER BY avg_salary DESC
LIMIT 1;
```

```
SELECT
dept_id,
AVG(salary) -- AVG(salary) IS THE NEW COLUMN NAME
FROM employee
GROUP BY dept_id
ORDER BY AVG(salary) DESC
LIMIT 1;
```

-- RANK BASED APPROACH

-- What is the issue with this code ?

```
-- ❌ SELECT *  
-- FROM (  
-- SELECT  
-- e.dept_id,  
-- AVG(e.salary) AS avg_salary,  
-- DENSE_RANK() OVER(ORDER BY avg_salay DESC) AS rnk  
-- FROM employee e  
-- GROUP BY e.dept_id  
-- ) x  
-- ❌ WHERE rnk=1;
```

```
SELECT *  
FROM (  
SELECT  
e.dept_id,  
AVG(e.salary) AS avg_salary,  
DENSE_RANK() OVER(ORDER BY AVG(e.salary) DESC) AS rnk  
FROM employee e  
GROUP BY e.dept_id  
) x  
WHERE rnk=1;
```

Find the highest salary in each department

use sql_practice_2026;

-- Date: Oct 3, 2026 | Time: 6:25:06 PM | MYSQL 8.0+

/* Possible Approaches :

*/

-- SQL Question Find the highest salary in each department:

-- GROUP BY

SELECT dept_id , MAX(salary) AS heighest_salary

FROM employee

GROUP BY dept_id;

Find the maximum salary without using MAX

use sql_practice_2026;

-- Date: Oct 3, 2026 | Time: 11:30:15 PM | MYSQL 8.0+

-- SQL Question : Find the maximum salary without using MAX

SELECT DISTINCT * FROM employee

ORDER BY salary DESC

LIMIT 1;

Find the Nth highest salary in each department

use sql_practice_2026;

-- Date: Oct 5, 2026 | Time: 2:46:08 PM | MYSQL 8.0+

-- SQL Question : Find the Nth highest salary in each department

SELECT *FROM* (

SELECT

e,

DENSE_RANK() OVER(PARTITION BY dept_id ORDER BY salary DESC) AS rn

FROM employee e

) x

WHERE rn = 2;

Find the Nth highest salary without LIMIT OR TOP

use sql_practice_2026;

-- Date: Oct 3, 2026 | Time: 11:31:36 PM | MYSQL 8.0+

-- SQL Question : Find the Nth highest salary without LIMIT OR TOP.

-- DENSE RANK is coming to my mind

SELECT

FROM

(

SELECT

e,

DENSE_RANK() OVER(ORDER BY salary DESC) AS rnk

FROM employee e

) x

WHERE rnk=2;

Find the Nth highest salary

use sql_practice_2026;

-- Date: Oct 3, 2026 | Time: 6:18:58 PM | MYSQL 8.0+

/* Possible Approaches :

*/

-- SQL Question Find the Nth highest salary:

SELECT *FROM* (

SELECT

e,

RANK() OVER(ORDER BY salary DESC) AS rnk -- Remember it

FROM employee e

) x -- This is Important

WHERE rnk=2;

Find the salary difference between employee and manager

use sql_practice_2026;

-- Date: Oct 5, 2026 | Time: 2:57:08 PM | MYSQL 8.0+

-- SQL Question : Find the salary difference between employee and manager.

SELECT * FROM employee;

-- Thinking of Join

```
SELECT
e.salary ,
m.salary ,
(e.salary - m.salary) AS diff
FROM employee e
JOIN employee m
ON e.manager_id = m.employee_id;
```

```
SELECT
e.salary as emp_salary,
m.salary as manager_salary,
ABS(e.salary - m.salary) AS diff
FROM employee e
JOIN employee m
ON e.manager_id = m.employee_id;
```

Find the second highest salary in each department.

use sql_practice_2026;

-- Date: Oct 3, 2026 | Time: 11:06:41 PM | MYSQL 8.0+

-- SQL Question : Find the second highest salary in each department.

SELECT * FROM employee ;

SELECT

FROM (

SELECT

e,

DENSE_RANK() OVER(PARTITION BY dept_id ORDER BY salary DESC) AS rnk

FROM employee e

) x

WHERE rnk=2;

Find the top 3 salaries in each department.

use sql_pratice_2026;

-- Date: Oct 3, 2026 | Time: 6:28:36 PM | MYSQL 8.0+

/* Possible Approaches :

*/

-- SQL Question Find the top 3 salaries in each department.:

-- GROUP BY | Reason : It combine the result

SELECT dept_id, MAX(salary) AS salary FROM employee

GROUP BY dept_id

ORDER BY

dept_id ASC,

salary DESC ;

-- MAX() → gives only the highest salary,

-- not the top 3 salaries.

-- Can we solve by GROUP ? NOOOOOOOO

-- Approach 2: DENSE_RANK()

SELECT *FROM*

(

SELECT

e,

DENSE_RANK() OVER(PARTITION BY dept_id ORDER BY salary DESC) AS rnk

FROM employee e

) x

WHERE rnk <= 3

ORDER BY dept_id ASC, salary DESC;

Finds Managers Who Have More Than One Employee

use sql_practice_2026;

-- Date: Oct 5, 2026 | Time: 1:48:36 PM | MYSQL 8.0+

-- SQL Question : Finds Managers Who Have More Than One Employee.

SELECT * FROM employee;

-- finds managers who have more than one employee.

SELECT

manager_id,

COUNT() AS cnt

FROM employee

GROUP BY manager_id

HAVING COUNT()>1

;